

CHAPTER 5

Activities & Lab Packet

Forces in Nature

A complete set of labs, worksheets, writing prompts, discussions, and group projects to accompany Chapter 5. Works with both the Standard and Easy Read versions of the textbook.

 Georgia Standard S8P5

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1 Static Electricity Stations

LAB

OBJECTIVE

Observe charge transfer through friction, conduction, and induction.

TIME

40–45 minutes (rotating stations)

STANDARDS

S8P5.a, S8P5.b

Setup: 4 Station Lab

Set up four stations around the room. Groups rotate through each station in 8-10 minutes. Each station shows a different charging effect.

Station 1 — Balloon & Hair (Friction)

- 3–4 inflated balloons
- Small mirror (optional)

Instructions: Rub the balloon vigorously on your hair (or a wool sweater) for 10 seconds. Then SLOWLY pull it away. Watch what happens to your hair. Try sticking the balloon to the wall, ceiling, or door.

Station 2 — Bending Water (Induction)

- 1 sink or large bowl
- 1 plastic comb or PVC pipe (small section)
- Wool cloth or piece of fur

Instructions: Turn the faucet on a thin trickle. Rub the comb on the wool cloth for 15 seconds. Bring the charged comb CLOSE TO (but not touching) the stream of water. Watch the water bend!

Station 3 — Confetti Jump (Induction + Conduction)

- Tiny paper bits (hole-punch confetti works great)
- 1 inflated balloon or charged comb
- Wool cloth

Instructions: Place a small pile of paper confetti on a flat surface. Rub the balloon on wool. Hold the balloon a few inches above the paper. The paper bits should JUMP up to the balloon!

Station 4 — The Doorknob Shock (Conduction)

- Carpet area (or rug)
- Metal doorknob, faucet, or filing cabinet

Instructions: Shuffle your feet across the carpet (or rub your feet on a rug). Slowly bring your finger near a metal object. You should see (and feel!) a small spark — a transfer of electrons through conduction.

TEACHER NOTES

The science at each station:

- **Station 1:** FRICTION transfers electrons from hair → balloon. Hair becomes positive, balloon becomes negative. Like charges on each hair repel, so hair stands up.
- **Station 2:** The charged comb causes the water molecules (which have a slight + and – end) to ALIGN. The water bends toward the comb. INDUCTION!
- **Station 3:** The charged balloon INDUCES opposite charges on the paper surface. The paper gets attracted, jumps up, and may even bounce off after touching (conduction).
- **Station 4:** FRICTION builds charge on you. CONDUCTION dumps it into the doorknob as a spark.

Tip: Dry weather works MUCH better for static experiments. If it's humid, the effects will be weaker.

Safety: The doorknob shock is harmless but can surprise. Warn students!

Static Electricity Stations — Data Sheet

Name: _____

Period: _____

Station	What I Observed	Charging Method (friction, conduction, or induction?)
1. Balloon & Hair		
2. Bending Water		
3. Confetti Jump		
4. Doorknob Shock		

Why did the water bend at Station 2 (even though the comb never touched it)?

In which stations did electrons MOVE FROM one object to another?

Where did the energy come from to create the doorknob spark?

2 Magnet Exploration

LAB

OBJECTIVE

Investigate magnetic poles, field strength, and which materials attract.

TIME

30 minutes

STANDARDS

S8P5.a, S8P5.c

Materials (per group)

- 2 bar magnets (or 2 strong refrigerator magnets — neodymium magnets work great)
- Test materials: paperclip, penny (post-1982), nickel, aluminum foil, plastic spoon, wood pencil, rubber band, iron nail, steel washer, paper
- 1 piece of plain paper
- Iron filings (if available — or use small pieces of steel wool cut up)
- Compass (optional but cool)

Three Experiments

Experiment A — Pole Test

1. Hold one bar magnet still. Bring the second magnet's N pole toward its N pole. What happens?
2. Now flip the second magnet so the S pole approaches the first magnet's N pole. What happens?
3. Record which combinations REPEL and which ATTRACT.

Experiment B — Material Test

1. One by one, hold each test material near a magnet.
2. Sort each material into "attracted" or "not attracted."
3. What pattern do you notice about which materials are attracted?

Experiment C — See the Field

1. Place a piece of paper over a bar magnet.
2. Sprinkle iron filings (or tiny pieces of steel wool) lightly across the paper.
3. Tap the paper gently. Watch what shapes the filings form!
4. Sketch the field lines you see.

TEACHER NOTES

Expected results:

- **Pole test:** Like poles repel (N-N or S-S). Opposite poles attract (N-S).
- **Material test:** Only iron, nickel, cobalt, and steel are magnetic. Pennies (after 1982 are zinc + copper), aluminum, plastic, wood, and paper are NOT magnetic.
- **Field lines:** Iron filings line up along the magnetic field lines, showing the curved field that goes from N pole around to S pole.

Heads-up on pennies: Pre-1982 pennies are mostly copper (NOT magnetic). Post-1982 pennies are zinc-coated copper (also not magnetic). Neither will be attracted. This is good — it lets students realize "metal" doesn't always mean "magnetic."

Safety with neodymium magnets: If you use strong rare-earth magnets, warn students NOT to let two strong magnets snap together — they can pinch fingers hard, break, or send shards flying.

Magnet Exploration — Data Sheet

Name: _____

Period: _____

Experiment A: Pole Test

Pole Combination	Attract or Repel?
N pole + N pole	
S pole + S pole	
N pole + S pole	

Experiment B: Material Test (Write Y for attracted, N for not attracted)

Paperclip: _____

Penny: _____

Nickel: _____

Aluminum foil: _____

Plastic spoon: _____

Wood pencil: _____

Rubber band: _____

Iron nail: _____

Steel washer: _____

Paper: _____

Experiment C: Sketch the field lines you saw with the iron filings:

What pattern do you notice about which materials were attracted to the magnet?

3 Build an Electromagnet

LAB

OBJECTIVE

Build an electromagnet and test what makes it stronger.

TIME

35–45 minutes

STANDARDS

S8P5.c

Materials (per group)

- 1 large iron nail (3+ inches long, must be iron — not stainless steel)
- 1 meter of insulated copper wire (thin gauge works best)
- 1 D-cell battery (or AA — D works better)
- Optional 2nd D-cell battery
- Small paperclips (10–20)
- Masking tape

Setup

Strip about 1 inch of insulation off each end of the wire. (You may need to do this for the students.) Wrap the wire around the nail about 20 times, leaving the two stripped ends free.

Student Instructions

- 1. Build it:** Wrap the wire 20 times around the nail. Keep the coils tight and close together.
- 2. Connect it:** Touch one stripped end of the wire to the + side of the battery. Touch the other end to the – side. (Tape can help hold them.)
- 3. Test it:** Dip the nail into a pile of paperclips. How many does it pick up? Record.
- 4. Disconnect it:** Take the wire off the battery. The paperclips should fall off (or hold much weaker)!
- 5. Try variations:**
 - Add 10 more coils (so 30 total). Test again.
 - Add 10 more (40 coils). Test again.
 - Use 2 batteries connected end-to-end (in series). Test again.

TEACHER NOTES

The big concept: Three variables affect electromagnet strength:

- Number of coils — more coils = stronger
- Amount of current (more batteries) = stronger
- Type of core (iron is best)

This is exactly what Section 5.9 of the textbook covers (S8P5.c).

Safety: The wire gets HOT when connected. Don't leave it connected for more than 30 seconds at a time, and don't touch the wire while it's connected. Use tape to keep contact rather than fingers.

If it doesn't work: 99% of the time it's because the nail is stainless steel (not iron). Test with a regular magnet first — if a magnet sticks to the nail, the nail will work as a core.

Extension: Discuss real-world electromagnets — junkyard cranes, MRI machines, maglev trains, electric motors, speakers, doorbells.

Build an Electromagnet — Data Sheet

Name: _____

Period: _____

Results:

Setup	# of Coils	# of Batteries	# of Paperclips Picked Up
Trial 1 (basic)	20	1	
Trial 2 (more coils)	30	1	
Trial 3 (most coils)	40	1	
Trial 4 (more current)	40	2	
Disconnected battery	—	0	

What happened when you DISCONNECTED the battery? Why?

What made the electromagnet STRONGER? List the two variables you tested.

Give a real-world example of an electromagnet:

4 Mass vs. Weight Practice

WORKSHEET

OBJECTIVE

Distinguish between mass and weight and calculate weight on different planets.

TIME

15–20 minutes

STANDARDS

S8P5.a

Mass vs. Weight

Mass stays the same. Weight changes with gravity. Weight = Mass $\times$ Gravity.

Name: _____

Period: _____

Part A: Mass or Weight? (Circle one)

- "I weigh 110 pounds." → **Mass / Weight**
- "This dog has a mass of 25 kg." → **Mass / Weight**
- "The astronaut feels weightless on the space station." → **Mass / Weight**
- "A bag of sugar contains 2 kg of sweet stuff." → **Mass / Weight**
- "On the Moon, you would feel only 1/6 as heavy." → **Mass / Weight**

Part B: True or False

- Your mass changes when you travel to the Moon. _____
- Weight is a measure of how hard gravity pulls on you. _____
- Newtons is a unit of mass. _____
- You would weigh MORE on Jupiter than on Earth. _____
- Astronauts on the ISS are weightless because there's no gravity. _____

Part C: Use the table to answer the questions.

Place	Gravity	Your mass (kg)	Your weight (N)
Earth	9.8 m/s ²	50 kg	490 N
Moon	1.6 m/s ²	50 kg	80 N
Mars	3.7 m/s ²	50 kg	185 N
Jupiter	24.8 m/s ²	50 kg	1,240 N

- Where would you feel HEAVIEST? _____
- Where would you feel LIGHTEST (still on a surface)? _____
- What stays the SAME on all four planets? _____
- What CHANGES between the four planets? _____

ANSWER KEY

Part A: 1) Weight (pounds = unit of weight) | 2) Mass (kg = unit of mass) | 3) Weight | 4) Mass | 5) Weight

Part B: 6) F (mass never changes) | 7) T | 8) F (newtons measure force/weight, NOT mass) | 9) T | 10) F (there IS gravity — they're in free-fall around Earth)

Part C: 11) Jupiter | 12) Moon | 13) Mass (50 kg everywhere) | 14) Weight (and gravity)

5 Charging Methods ID

WORKSHEET

OBJECTIVE

Identify friction, conduction, and induction in real-world scenarios.

TIME

15 minutes

STANDARDS

S8P5.b

Three Ways to Charge — Identify Each One

Write F for Friction, C for Conduction, or I for Induction.

Name: _____

Period: _____

FRICTION	CONDUCTION	INDUCTION
rubbing two things together	objects touching	close, but NOT touching

#	Scenario	F, C, or I?
1	You rub a balloon on your hair and it becomes negatively charged.	
2	A charged balloon held NEAR a wall sticks to the wall without rubbing it.	
3	You touch a metal doorknob after shuffling on carpet — ZAP!	
4	Static cling makes clothes stick together right out of the dryer.	
5	A charged comb causes small bits of paper to JUMP up to it without touching them first.	
6	A charged ball touches a neutral ball — now both have a charge.	
7	You take off a wool sweater and your hair stands up.	
8	A charged rod is held near a faucet stream — water bends toward it.	
9	Two metal spheres touch and share the charge equally.	
10	Plastic wrap clings to a glass bowl because of how it was pulled off the roll.	

Bonus: In INDUCTION, does the neutral object actually gain or lose electrons overall? Explain.

ANSWER KEY

1) F | 2) I | 3) C | 4) F | 5) I | 6) C | 7) F | 8) I | 9) C | 10) F

Bonus: No! In induction, no electrons are gained or lost overall. The charges INSIDE the neutral object just rearrange (the same-charge ones get pushed away, opposite-charge ones get pulled toward the charged object). The total charge stays zero.

6 Conductor or Insulator?

WORKSHEET

OBJECTIVE

Classify materials as conductors or insulators and apply to real situations.

TIME

15 minutes

STANDARDS

S8P5.b

Conductor or Insulator?

Conductors let electricity flow easily. Insulators block the flow.

Name: _____

Period: _____

Part A: Classify each material. Write C for Conductor or I for Insulator.

- | | |
|----------------------------|-------------------------|
| 1. Copper wire: _____ | 2. Rubber glove: _____ |
| 3. Wood spoon: _____ | 4. Aluminum foil: _____ |
| 5. Glass cup: _____ | 6. Iron nail: _____ |
| 7. Plastic ruler: _____ | 8. Silver coin: _____ |
| 9. Dry air: _____ | 10. Salt water: _____ |
| 11. Steel paperclip: _____ | 12. Ceramic mug: _____ |

Part B: Apply It

1. Why are electrical wires made of **copper** inside but coated with **plastic** on the outside?

2. Why is it dangerous to use a hair dryer near a bathtub full of water?

3. Why do electricians wear rubber-soled boots and rubber gloves?

Part C: Charge Distribution

On a CONDUCTOR, charges **SPREAD OUT / STAY PUT** across the surface.

On an INSULATOR, charges **SPREAD OUT / STAY PUT** where they were placed.

ANSWER KEY

Part A: 1) C | 2) I | 3) I | 4) C | 5) I | 6) C | 7) I | 8) C | 9) I | 10) C | 11) C | 12) I

Part B: 1) Copper conducts electricity well (current flows through the wire). Plastic insulates — it keeps electricity from jumping out into your hand. | 2) Water (especially with minerals/salts) is a conductor. Electricity could flow from the dryer through the water and through you. | 3) Rubber is an insulator. It stops electricity from flowing through their body to the ground.

Part C: Conductors → SPREAD OUT | Insulators → STAY PUT

7 Writing Prompts

WRITING

OBJECTIVE

Use scientific vocabulary to explain forces in nature concepts.

TIME

15–30 minutes each

STANDARDS

Cross-cutting

Level 1 — Beginner (1 paragraph)

1. **Heaviest Day Ever.** Imagine you woke up on Jupiter. Describe what your day would feel like with gravity 2.5 times stronger than Earth. What would be hard? What would still work?
2. **Static Surprise.** Write about a time you got "zapped" by static electricity. Now explain what caused it using the words "friction," "electrons," and "conduction."
3. **Magnet Hunt.** Look around the room you're in. Name three things that contain a magnet or use magnetism. Explain how each one uses magnets.

Level 2 — Intermediate (3 paragraphs)

4. **The Lightning Story.** Lightning is just static electricity at a HUGE scale. Trace how lightning forms inside a thunderstorm cloud. Use friction, electrons, and conduction in your explanation.
5. **Mass, Weight, and the Moon.** An astronaut weighs 540 N on Earth but only 90 N on the Moon. Explain why the weight changed but the astronaut's mass did NOT.
6. **Three Forces, One World.** Earth has three big non-contact forces shaping it: gravity, electric, and magnetic. Pick ONE and explain how the world would be different if it suddenly stopped existing.

Level 3 — Advanced (essay)

7. **The Electromagnet Argument.** Some scientists call the discovery that electricity and magnetism are connected one of the most important breakthroughs in history. Defend that claim. How does this connection power our modern world? Use specific examples (motors, generators, power plants, etc.)
8. **Without Earth's Magnetic Field.** Earth's molten iron core gives it a magnetic field that protects us from the Sun's harmful radiation. Imagine that field disappeared overnight. Write a creative essay describing what would happen to life on Earth over the following weeks, months, and years.

8 Classroom Discussion Questions

DISCUSSION

OBJECTIVE

Practice scientific thinking through verbal reasoning.

USE

Warm-ups, exit tickets, Socratic seminar

STANDARDS

Cross-cutting

TEACHER NOTES

Use think-pair-share format. The "deeper" questions are great for sparking debate. Push students to defend with evidence from the chapter, not just guesses.

Quick Warm-Ups (1–2 min each)

1. Why don't you feel the gravity between you and your desk?
2. Can two North poles ever stick together? Why or why not?
3. What's the difference between mass and weight in one sentence?
4. Why does a balloon stick to a wall after you rub it on your hair?
5. If you turn off the electricity to an electromagnet, what happens?

Deeper Discussions (5–15 min each)

6. **Bird on a Wire.** Why doesn't a bird sitting on a high-voltage power line get electrocuted? What **WOULD** electrocute the bird? Use what you know about conductors, circuits, and current flow.
7. **The ISS Paradox.** Astronauts on the International Space Station appear "weightless," yet Earth's gravity is still pulling on them with almost 90% the force it does on the ground. How can both be true?
8. **Why Compasses Work.** Compass needles always point north. What makes this happen — and what would happen to a compass on Mars (which has no magnetic field)?
9. **Touchscreens vs. Buttons.** Why does a touchscreen respond to your finger but **NOT** to a pencil eraser? Tie your answer to induction and conductors/insulators.
10. **The Power Grid.** Discuss: every time you flip a light switch, somewhere a massive magnet is spinning inside a power plant. Explain the chain of physics from that spinning magnet to your light bulb.

9 Junkyard Crane Challenge

GROUP PROJECT

OBJECTIVE

Design the strongest electromagnet using one of the four variables from S8P5.c.

TIME

2–3 class periods

STANDARDS

S8P5.c

Project Description

Each group designs and builds an electromagnet to pick up the most "scrap metal" possible — just like a real junkyard crane. Groups compete to design the strongest electromagnet using **ONLY** the materials provided and the variables they've learned about.

The Variables Students Can Control

Variable	How to change it
Number of coils	Wrap the wire more or fewer times
Amount of current	Use 1, 2, or 3 batteries connected in series
Size of iron core	Use a smaller or larger iron nail/bolt
Type of core	Iron, steel, or different materials

Project Requirements

- The electromagnet must be able to be turned ON and OFF
- Students must do at least 3 different design tests (changing **ONE** variable each time)
- Groups must keep a data log of their tests
- Final design pulls up "scrap" (small paperclips, steel washers, or nails) in a class competition
- Groups must explain **WHY** their design is strong using the four variables

Easy-to-find Materials

- Insulated copper wire (1–3 meters per group)
- Iron nails (3–6 inches long), bolts, or carriage bolts
- D-cell or AA batteries (1–3 per group)
- Tape, rubber bands
- Test cargo: paperclips, steel washers, small nails, BBs

Required Deliverables

- 1. Design log** — record each test variation and how many paperclips it lifted.
- 2. Final design** — the strongest version, ready for the competition.
- 3. Written explanation** — Which variable made the **BIGGEST** difference? Which made the least? Why?
- 4. Class competition** — each group's electromagnet competes to lift the most paperclips in 30 seconds.

TEACHER NOTES

What students should learn: All four variables affect electromagnet strength. The biggest impact usually comes from more coils + more current. This directly assesses S8P5.c.

Safety: Limit how long the wire can stay connected (30 seconds max). The wire WILL get hot. Watch for kids who connect batteries directly without a coil — that's a short circuit and can heat up dangerously.

Tip: Pre-strip the wire ends for groups to save class time.

Rubric (suggested): Design works (25%), design log filled out (25%), written explanation accuracy (25%), competition performance & teamwork (25%).

Real-world tie-in: Show a video clip of a real junkyard crane in action. Students love seeing this in real life — it makes the concept stick.

10 Forces in Nature Jeopardy

REVIEW GAME

OBJECTIVE

Review chapter vocabulary and concepts in a competitive game format.

TIME

30–40 minutes

STANDARDS

S8P5 cross-cutting

How to Play

Divide the class into 2–4 teams. Use the question board below (you can copy it onto chart paper, a whiteboard, or a slide). Teams take turns picking a category and point value. Read the answer/clue aloud. The team must respond in the form of a question. If they're right, they earn the points. If wrong, the next team can steal.

Game Board — Five Categories, Five Point Values

GRAVITY	ELECTRIC CHARGE	CONDUCTORS & INSULATORS	MAGNETS	ELECTROMAGNETS
\$100: The force that pulls objects with mass toward each other.	\$100: The particle with a NEGATIVE charge.	\$100: A material that lets electricity flow easily.	\$100: The two ends of a magnet are called these.	\$100: A magnet you can turn on and off using electricity.
\$200: This is measured in kilograms and never changes.	\$200: The particle with a POSITIVE charge.	\$200: A material that blocks electricity.	\$200: Like poles do this when brought together.	\$200: Wrapping more wire around an electromagnet does this to its strength.
\$300: Your weight on the Moon compared to Earth (in fractions).	\$300: This is what happens when you bring two negative charges together.	\$300: This is what charges do on a conductor's surface.	\$300: The planet that acts as a giant bar magnet.	\$300: Adding more batteries to an electromagnet does this.
\$400: The two factors that affect the strength of gravity between two objects.	\$400: The method of charging where two things rub together.	\$400: Why power lines have plastic on the outside but metal on the inside.	\$400: The lines that show the shape of a magnetic field.	\$400: What happens to an electromagnet when the battery is disconnected.
\$500: This is why astronauts on the ISS appear weightless even though gravity is still there.	\$500: The method of charging where one object affects another WITHOUT touching.	\$500: This is what charges do on an insulator's surface.	\$500: What protects Earth from the Sun's harmful radiation.	\$500: The big idea that links electricity and magnetism.

Answer Key for the Teacher

Cat. / \$	Correct Response (in question form)
GRAVITY \$100	What is gravity?
GRAVITY \$200	What is mass?
GRAVITY \$300	What is 1/6 (one-sixth)?
GRAVITY \$400	What are mass and distance?
GRAVITY \$500	What is free-fall? / What is being in orbit?
CHARGE \$100	What is an electron?
CHARGE \$200	What is a proton?
CHARGE \$300	What is repel (or push apart)?
CHARGE \$400	What is friction?
CHARGE \$500	What is induction?
COND/INS \$100	What is a conductor?
COND/INS \$200	What is an insulator?
COND/INS \$300	What is spread out evenly?
COND/INS \$400	What is the metal conducts electricity (and the plastic insulates / protects)?
COND/INS \$500	What is stay put (where they were placed)?
MAGNETS \$100	What are poles? (or North and South)
MAGNETS \$200	What is repel? (push apart)
MAGNETS \$300	What is Earth?
MAGNETS \$400	What are magnetic field lines?
MAGNETS \$500	What is Earth's magnetic field?
ELEC-M \$100	What is an electromagnet?
ELEC-M \$200	What is increases? (makes it stronger)
ELEC-M \$300	What is makes it stronger? (increases the strength)
ELEC-M \$400	What is the magnet turns off?
ELEC-M \$500	What is electromagnetism? (or moving charges make magnetic fields and vice versa)

TEACHER NOTES

Setup tips: Project the board on the screen or write it on chart paper. As teams answer, cross out the squares they pick.

Daily Doubles: Mark 2–3 squares in advance as "Daily Doubles" — the team that picks them can wager any amount of their current points.

Final Jeopardy: End with one big question — "Name and describe ALL THREE methods of charging an object." Teams wager any amount of their points.

Easier adaptation: Allow teams to give the answer in regular form (not "What is...?") if they're struggling with the format.

Quick prizes: Homework pass, candy, classroom job of choice for the winning team.